



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

April 11, 2023

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 11186
(TWENTY FIRST REVISION)

EXPIRATION DATE: 2027-02-28

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: Chart Inc.
New Prague, MN
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, marking, sale and use of a non-DOT specification vacuum insulated portable tank conforming with all requirements applicable to a DOT Specification MC 338 cargo tank motor vehicle for the transportation in commerce of the materials authorized by this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a) party status may not be granted to a manufacturing permit. These packaging may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.318 in that portable tanks are not authorized; §§ 176.30 and 176.76(g), except as specified herein.
5. BASIS: This special permit is based on Chart Inc.'s application dated March 23, 2023, submitted in accordance with § 107.109.

Tracking Number: 2023035550

6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Argon, refrigerated liquid (<i>cryogenic liquid</i>)	2.2	UN1951	N/A
Carbon Dioxide, refrigerated liquid	2.2	UN2187	N/A
Ethane, refrigerated liquid	2.1	UN1961	N/A
Ethylene, refrigerated liquid (<i>cryogenic liquid</i>)	2.1	UN1038	N/A
Helium, refrigerated liquid (<i>cryogenic liquid</i>)	2.2	UN1963	N/A
Methane, refrigerated liquid (<i>cryogenic liquid</i>) or Natural gas, refrigerated liquid (<i>cryogenic liquid</i>), with high methane content	2.1	UN1972	N/A
Nitrogen, refrigerated liquid (<i>cryogenic liquid</i>)	2.2	UN1977	N/A
Nitrous Oxide, refrigerated liquid	2.2	UN2201	N/A
Oxygen, refrigerated liquid (<i>cryogenic liquid</i>)	2.2	UN1073	N/A

7. SAFETY CONTROL MEASURES:

a. PACKAGING: Prescribed packagings are 12 models of non-DOT specification portable tanks designed, constructed and “U” stamped in accordance with Section VIII, Division 1 of the ASME Code. Each tank must conform to the design criteria set forth below:

CRYENCO MODEL NUMBER	WORKING PRESSURE (PSIG)	MINIMUM TEMP.(°F)	VOLUME (GALS)	DRAWING NUMBER
TVS-23-PB-45	45	-320	2300	257794
TVS-33-PB-45	45	-320	3306	254967 or 258778
TVS-22-PB-90	90	-320	2283	253005
TVS-32-PB-90	90	-320	3289	253856-10
TVS-54-PB-60	60	-320	5400	252363
TVS-54-VB-60	60	-320	5400	253191
TVS-53-PB-150	150	-320	5350	118467
TVS-53-VB-150	150	-320	5350	253763
TVS-53-B-150-He	150	-453	5200	252846
TVS-52-PB-250	250	-320	5270	251966
TVN-52-EB-350	350	-320	5200	256468
TVN-520B--350	350	-320	5200	256098

Each portable tank is vacuum insulated and enclosed in a frame that meets all requirements of an ISO standard frame except for overall dimensions. The portable tank must conform to Chart, Inc. (formerly Cryenco, Inc.'s) drawings, calculations and specifications on file with the Office of Hazardous Materials Safety (OHMS). Packagings authorized must conform with § 178.338 except as follows:

- (1) § 178.338-2(c): Impact testing is not required for stainless steels used for a lading warmer than -425 F.

(2) § 178.338-6(b): Each portable tank in oxygen or nitrous oxide service must be provided with an inspection access hole (manhole) of not less than 16.0 inches (406 mm) diameter. After a final inspection the access hole must be closed by welding using a suitable access cover plate fabricated from the same material as the tank. The tank must be provided with a means of entrance and exit through the jacket, or the jacket must be marked to indicate the access hole location.

(3) § 178.338-10: This section does not apply.

(4) § 178.338-13(a): Lifting lugs, framework and any anchoring to the inner tank or the tank jacket must conform with § 178.338-13(a). The portable tank need not conform to § 178.338-13(b) or (c).

(5) § 178.338-18(b)(1): Each portable tank must be plainly and durably marked "DOT-SP 11186" in place of the DOT Specification Number MC-338.

(6) A portable tank that meets the definition of "container" must meet the requirements of 49 CFR parts 450 through 453, and each design must be qualified in accordance with § 178.270-13(c).

b. TESTING: The portable tank must be reinspected and retested once every five years in accordance with the procedure prescribed in § 180.605(g) for DOT Specification 51 portable tanks. In place of the requirement for visual inspection, before and after vacuum readings must be used to detect leakage. Nitrogen or an inert gas may be used as a test medium in place of air or water as required by § 180.605(h)(3). The test pressure for the inner tank must be determined from the following formula:

$$P_T = 1.25 \times [P_d] - 14.7$$

Where:

P_T = Test pressure, psig

P_d = Design pressure

(the sum of the maximum allowable working pressure, liquid head and 14.7 psi)

c. OPERATIONAL CONTROLS:

(1) Each portable tank must be prepared and shipped as required in § 173.318, as applicable for the lading.

(2) No person may transport a portable tank containing Division 2.1 liquid unless the pressure of the lading is equal to or less than that used to determine the marked rated holding time and the OWTT is equal to or greater than the elapsed time between the start and termination of travel.

(3) For the transport of Division 2.1 liquid, the actual holding time for each tank must be determined after each shipment. If it is determined that the actual holding time is less than 90 percent of the MRHT of the tank, the tank may not be refilled until it is restored to its MRHT, or the tank is remarked with the reduced holding time determined by this examination.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.

c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

d. Each packaging manufactured under the authority of this special permit must be either (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured or (2) marked with a registration symbol designated by the Office of Hazardous Materials Safety for a specific manufacturing facility.

e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.

f. Each portable tank must be plainly marked on both sides near the middle, in letters at least two inches high on a contrasting background, "DOT-SP 11186", in place of "MC 338".

g. Transportation of Division 2.1 (flammable gases) materials are not authorized aboard cargo vessel unless specifically authorized in the Hazardous Materials Table (§ 172.101).

- h. New construction is not authorized.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, and cargo vessel.
10. MODAL REQUIREMENTS:
- a. A current copy of this special permit must be carried aboard each cargo vessel or motor vehicle used to transport packages covered by this special permit.
 - b. Shipments by cargo vessel must conform with the following:
 - (1) The package and its stowage must conform with § 176.76(g). In all situations, the portable tanks must be stowed such that they are readily accessible and can be monitored in accordance with the provisions of this special permit. Portable tanks may be overstowed only if enclosed in ISO frames and the following provisions are met:
 - (i) The pressure of the lading is equal to or less than that used to determine the marked rated holding time and the OWTT is equal to or greater than the elapsed time between the start and termination of travel.
 - (ii) The actual holding time for each tank must be determined after each shipment. If it is determined that the actual holding time is less than 90% of the MRHT of the tank, a charged tank may not be overstowed until it is restored to its MRHT, or the tank is re-marked with the reduced holding time determined by this special permit.
 - (2) The legend "One-Way Travel Time Hours" must be marked on the shipping paper or on the dangerous cargo manifest immediately after the container description. The OWTT is determined by the formula:
$$\text{OWTT} = \text{MRHT} - 24 \text{ hours.}$$
 - (3) Any lading road relief valve set at a pressure lower than that prescribed for the (safety) pressure relief valve must be closed during transportation by cargo vessel.
 - (4) The requirements of paragraphs, 10.b(2) and (3) above are waived if all of the following conditions are met:
 - (i) The lading is liquid nitrogen.

(ii) Transportation by cargo vessel is to oil and gas production facilities within the jurisdiction of the United States of America.

(iii) The portable tank is not overstowed with other containers or freight.

c. The portable tank may not be transported in container-on-flat car (COFC) or trailer-on-flat car (TOFC) service except under conditions approved by the Associate Administrator for Safety, Federal Railroad Administration.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by Federal hazardous materials transportation law 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8 who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by § 172.700 through § 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)- 'The Hazardous Materials Safety and Security Reauthorization Act of 2005' (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term "exemption" to "special permit" and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

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